RAKESH BORRA

borrarakesh369@gmail.com | +1 617-602-3006 | LinkedIn | GitHub | Boston, MA | Portfolio

OBJECTIVE

MS Computer Science student with experience in software testing, debugging, API integration, database-driven applications, and technical problem-solving, seeking entry-level QA Automation, Application Support, or Implementation Engineer opportunities.

EDUCATION

University of Massachusetts Boston, MA, USA **Jan 2025 – Present**
Master of Science in Computer Science

Indian Institute of Information Technology Sricity, India
Bachelor of Technology in Computer Science and Engineering

SKILLS

- **Programming:** Python, JavaScript, SQL
- **Testing:** pytest, Unit Testing, Debugging, Data Validation
- **APIs & Databases:** REST APIs, Firebase, SQLite, PostgreSQL
- **Tools:** Git, GitHub, Docker

EXPERIENCE

Graduate Teaching Assistant – PHYS 113 **Sep 2026 – Present**
University of Massachusetts Boston

- Evaluate quantitative assignments and exams, identify errors, and provide consistent, detailed feedback.
- Explain technical problem-solving methods during office hours and coordinate grading with instructional staff.

Dining Services Associate **Jan 2026 – Aug 2026**
Sodexo – University of Massachusetts Boston

- Collaborated with team members to support time-sensitive operations and provide responsive service in a fast-paced environment.

PROJECTS

Clinical Trial AI – Protocol Analysis Platform (Python / Testing / Validation)

- Collaborated in a team of three to build a Python application for extracting and structuring clinical trial PDF information using PyMuPDF, OCR, and local LLMs.
- Developed source-page verification and automated testing workflows to identify missing information and support traceable, consistent protocol summaries.

Real-Time Real Estate Investment Analysis Web Application (React / Firebase / APIs)

- Developed authenticated application workflows using React, Firebase Authentication, and Firestore with user-scoped database rules.
- Integrated property-data APIs and implemented ROI, cash flow, and cap rate calculations with five-year projections.

Real-Time Smart Traffic Monitoring and Congestion Prediction System (FastAPI / PostgreSQL)

- Developed a traffic monitoring platform integrating YOLO, OpenCV, FastAPI, and React for real-time vehicle detection and congestion analysis.
- Integrated PostgreSQL-backed data workflows and dashboards to display traffic density and prediction results.

Project Pulse – Task Management Application (Kotlin / SQLite)

- Developed an Android task management application using Kotlin and SQLite with task tracking and persistent local storage.
- Implemented reminder scheduling using AlarmManager, task status indicators, and user login workflows.

CERTIFICATIONS

The Complete Data Structures and Algorithms Course in Python – Udemy | View Certificate